MUHAMMAD GULAM SYARIF

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Bachelor of Computer Engineering from Universitas Brawijaya with a focus on robotics, Internet of Things (IoT), and embedded systems, particularly humanoid robots. Experienced in developing ROS-based systems as well as web applications using Laravel and React. Proficient in Python and C++ for robotics applications and data processing. Highly adaptable, with strong problem-solving skills and the ability to work effectively in team environments.

EDUCATION

Brawijaya Universitas – Malang, Jawa Timur july 2022 – jan 2026

Bachelor Degree in Computer Engineering, 3.94/4.00

- Organisasi: Member of [Robotiik Filkom Ub] – [2025]
- Proyek: Research on humanoid robotics, including the development of the Universitas Brawijaya mascot robot.

EXPERIENCE

ROBOTIIK, Fakultas Ilmu Komputer, Universitas Brawijaya – Malang Jan 2023 – dec 2025

Divisi Elektrikal and Programing

- Developed and tested robot control and motion behaviors using Robot Operating System (ROS1 Noetic and ROS2 Jazzy).
- Co-developing a humanoid mascot robot for Universitas Brawijaya, including electronics design, actuator integration, and ROS-based motion control.
- Implemented a PID-based Zero Moment Point (ZMP) balance control system for the OP3 humanoid robot, integrating inverse kinematics to maintain stability during standing and walking.
- Designed and fabricated a custom button PCB for the OP3 humanoid robot using EasyEDA, from schematic to layout, achieving 100% functionality with zero hardware errors.
- Performed robot assembly and integration, ensuring all electrical and mechanical components operated reliably.

Robotic beginner, Universitas Brawijaya – Malang Oct 2024 – Nov 2024

Special Instructor – Electrical and Programming Division, Robin 2024

- Delivered hands-on training sessions on robot electronics, PCB design, microcontroller programming, and ROS (Robot Operating System) integration.
- Guided participants through practical implementation of robot control systems, ensuring they gained applied skills and technical understanding.
- Collaborated with fellow instructors to develop teaching materials and manage workshops throughout the event.

PKKMB dan Startup Academy, Universitas Brawijaya – Malang Aug 2024 - Nov 2024

Staff Divisi Disiplin Mahasiswa (Disma)

- Ensure that all participants comply with the established rules, such as punctual attendance, appropriate clothing, and good etiquette during the event. Follow up on participants who are late or absent without a clear reason, and provide sanctions in accordance with applicable regulations.

KRI, Universitas Muhammadiyah Surakarta – Surakarta may 2024 - July 2024

Finalis Divisi Robot Sepak Bola Humanoid – Kontes Robot Indonesia (KRI) 2024

- Successfully became a national finalist in the humanoid soccer robot category.
- Responsible for the robot electronic control system using ROS and microprocessor programming.

PKKMB dan Startup Academy, Universitas Brawijaya – Malang Aug 2023 - Nov 2023

Staff Divisi Disiplin Mahasiswa (Disma)

- Ensure that all participants comply with the established rules, such as punctual attendance, appropriate clothing, and good etiquette during the event. Follow up on participants who are late or absent without a clear reason, and provide sanctions in accordance with applicable regulations.

PROJECT

PID-Based Zero Moment Point (ZMP) Balance Control for Humanoid Robot

- Developed a balance control system that computes the robot's ZMP in real time and regulates it to a 0 cm setpoint using a PID controller.
- Integrated the controller with the robot's inverse kinematics so that changes in ZMP are compensated by adjusting body posture and joint positions during both standing and walking, improving overall robot stability.

Humanoid Robot Development: Universitas Brawijaya Mascot Robot

- Currently designing and developing a humanoid robot as the official mascot for Universitas Brawijaya, with 75% of the robot's framework completed and advanced robotic control systems being integrated using ROS.

Zero Moment Point (ZMP) Implementation on Humanoid Robot OP3

- Developed a dynamic stability analysis system using ZMP and Center of Mass (CoM) calculations for the OP3 humanoid robot.

OP3 Humanoid Robot Button PCB Design

- Designed and implemented the electrical button circuit for the OP3 humanoid robot as a custom PCB using EasyEDA, including schematic capture, layout, and fabrication-ready Gerber files.

Wearable Device for Stress Detection Using PPG Signal

- Developed a wearable device using MAX30102 and ESP32-S3 to detect stress by extracting SDNN and AVNN features from PPG signals, then classifying stress levels with 95% accuracy using the Random Forest algorithm.

Heart Disease Detection Using Support Vector Machine Algorithm

- Developed a machine learning model using the Support Vector Machine algorithm, achieving an accuracy of 0.852 on training data.

Drug Type Classification Using k-Nearest Neighbors Algorithm

- Built a classification system for drug type determination using the k-Nearest Neighbors algorithm with K=3, achieving an accuracy of 0.82.

Village Website Development: KKN Universitas Brawijaya

- I independently developed a Village Website using Laravel 11, Bootstrap, and Livewire.

SKILLS

- ROS (Robot Operating System): Familiar with developing robotics applications using ROS, including sensor integration and robot control.
- PCB Design: Skilled in designing printed circuit boards (PCBs) using industry standard Easy EDA software for electronic projects.
- Embedded Systems: Experienced in developing firmware and real-time applications using ESP32-S3 microcontroller, including sensor integration and wireless data transmission for wearable devices.
- Python: Proficient in Python for data analysis, signal processing, visualization, and robotics applications, including use of libraries such as Matplotlib, NumPy, and Tkinter.
- Laravel: Proficient in using Laravel for web development, building dynamic and secure web applications.
- Machine Learning: Experience in developing machine learning models and algorithms for data analysis and automation.
- C++: Strong knowledge of C++ for software development, with a focus on efficiency and performance optimization.
- PHP: Expertise in PHP for server-side development, creating robust and scalable web solutions.
- HTML/CSS: Proficient in front-end web development, creating responsive and user-friendly web pages with HTML and CSS.
- React.js: Developing interactive and responsive user interfaces using component-based architecture.
- Vue.js: Building modern front-end applications with reactive data binding and component-based design.